



Design and modeling of pipeline welding robot based on Pieper criterion and construction of integrated intelligent welding system

Chuanhui Zhu¹ · Zihao Wang¹ · Zhiming Zhu^{1,2} · Jichang Guo³

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Abstract

The welding efficiency and weld formation quality of pipeline external welding are important factors that constrain the site laying speed and service performance of long-distance oil and gas pipeline. It is urgent to optimize the structure and function of pipeline welding equipment and control system, and improve the intelligent level of the external welding process. This paper focuses on improving weld bead formation quality during the pipeline all-position welding. Based on the Pieper criterion, a 5-degree of freedom (DOF) welding robot is designed. It can meet the control requirements of welding torch position and posture (WTPP) that comply with the welding process parameters at any welding position of pipeline. The kinematic model of the designed robot is established, and an analytical solution to its inverse kinematics is derived. Combined with the composite sensor based on combined laser line structured lights vision sensor and gravity sensor, the pipeline all-position intelligent welding system is integrally constructed. Based on this, the detection method of welding groove size and WTPP parameters is proposed. Through the designed program control strategy, the intelligent adjustment experiments of the WTPP were carried out during the pipeline all-position welding process. The lateral tracking deviation of welding torch was not more than 0.25 mm; the vertical tracking deviation was not more than 0.63 mm, and the posture angles feedback control deviation were not more than 0.8°. The good weld formation effect was obtained, which provides support for the improvement of the intelligent level of pipeline external welding.

Keywords 5-DOF welding robot · Kinematic model · Composite sensing · Welding torch position and posture · Pipeline intelligent welding system

1 Introduction

Long-distance oil and gas pipeline has been an important national energy infrastructure for oil and gas transportation and supply, which is closely related to national economic

development and people's well-being. With the ongoing restructuring of the global energy system and the implementation of carbon neutrality strategies, the demand for clean energy sources such as natural gas continues to grow [1]. Consequently, oil and gas pipeline networks are expected to evolve toward higher steel grades and larger flow capacities, placing greater demands on the speed and quality of pipeline construction [2].

The external welding efficiency and weld formation quality of pipeline annular butt joints are key factors limiting pipeline construction speed and service performance. At present, all-position automatic external welding of annular butt joints at the pipeline laying site is generally carried out by external welding robots. The external welding process mainly includes thermal welding, filling, and cover welding, as shown in the weld bead filling for the double V-groove in Fig. 1. Due to the problems such as the deviation of groove machining at the end of pipeline section, the inconsistent assembly gap of pipeline section, the welding induced

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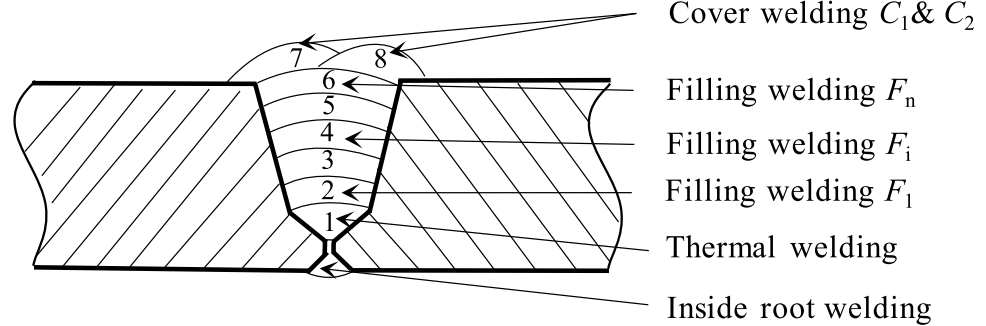
✉ Zhiming Zhu
zzmdme@mail.tsinghua.edu.cn

¹ Department of Mechanical Engineering, Tsinghua University, Beijing 100084, China

² Key Laboratory for Advanced Materials Processing Technology, Ministry of Education, Beijing, China

³ Coal Mining Branch, China, Coal Research Institute, Beijing 100013, China

Fig. 1 Diagram of weld bead filling for double V-groove



thermal deformation and so on, it is necessary to adjust the positions and angle of welding torch relative to the weld bead in real time, so as to ensure the stability of welding process, the accuracy of weld bead formation position, and reduce or even avoid the occurrence of weld defects [3]. In addition, due to the all-position welding of pipeline annular butt joints is a continuous welding process from flat welding to vertical welding and then to overhead welding, the molten pool at the vertical welding and the overhead welding positions may exhibit downward flow behavior due to gravity [4], leading to instability of the molten pool and affecting the weld formation quality [5]. It can be seen that the posture angle adjustment of the welding torch will have an important impact on the weld formation quality. Therefore, based on the welding position of the pipeline, by adjusting the torch tilt angle relative to the welding direction, the arc force can partially counteract the gravity-induced downward flow of the molten pool, thereby improving the weld formation quality during the pipeline all-position welding process [6, 7]. Furthermore, when laying pipeline on the slope such as hilly area, it is also necessary to properly set the torch oscillation angle relative to the groove side wall according to the pipeline laying angle and groove sizes, so as to obtain a good weld bead shape. However, the current pipeline external welding robots often lack the degrees of freedom (DOF) to adjust the posture of the welding torch [8]. This makes them unable to flexibly control the welding torch's posture during the all-position welding process. At the same time, in order to improve the weld formation quality of pipeline butt joints, reduce the labor intensity of workers, and enhance the automation and intelligent level of the welding process, online sensing detection technology is adopted in the pipeline welding process to provide real-time feedback control for pipeline welding robots, which is an important direction for further development in the welding field [9, 10]. Therefore, it is necessary to design a pipeline welding robot that can meet the requirements of flexibly adjusting the welding torch position and posture (WTPP) and utilize the online sensing system to detect the parameters required for effectively adjusting the WTPP and thus achieving intelligent welding [11].

Luo et al. [8] and Zhou et al. [12] separately designed the walking mechanism and the welding torch adjustment mechanism for the track-type pipeline welding robot, but the designed welding torch adjustment mechanism only has 3-DOF for adjusting the forward, lateral, and vertical positions of the welding torch, achieving weld tracking during pipeline welding. In order to meet the control requirements of welding torch DOF for pipeline all-position welding process and weld formation, Li et al. [13] designed a 4-DOF welding torch adjustment mechanism for welding robot. It can additively achieve the adjustment of the torch tilt angle, providing a basis for adjusting the welding torch to generate beneficial molten pool force field using different tilt angles in pipeline all-position welding process. Cao et al. [14] added a welding torch oscillation mechanism to the pipeline welding robot, which can realize the angular oscillation of the welding torch and edge stop during welding. However, neither of them comprehensively considers the need for WTPP control, meaning they cannot meet the requirement for arbitrary adjustment of WTPP when performing pipeline all-position welding on the slope.

The implementation of intelligent pipeline welding relies on the application of online sensing. Most of the information for welding process control by welders comes from visual perception. Active vision sensing based on laser line structured light has the advantages of simple image processing, high detection accuracy and applicable convenience. It has been widely used in the fields such as weld seam tracking, welding groove size parameter detection, welding torch relative position, and posture detection [15–19]. Combined with laser vision sensing, the trackless welding robot designed by Feng et al. [20] achieved seam tracking and automatic adjustment of the welding torch in both lateral and vertical directions during the welding process of the spherical tank; Zhu et al. [21] designed a partitioned time–frequency–spatial image processing algorithm for the dual-line structured light vision sensing, achieving high-precision welding torch position control during pipeline butt joint welding process. Research on weld seam tracking based on line

structured light vision sensing [22, 23] mainly focuses on image processing. The aim is to improve the detection accuracy and anti-interference of image processing algorithms, so as to better achieve welding torch position adjustment. But there is little research on integrated detection of welding positions and postures of workpieces to be welded and control of the WTPP.

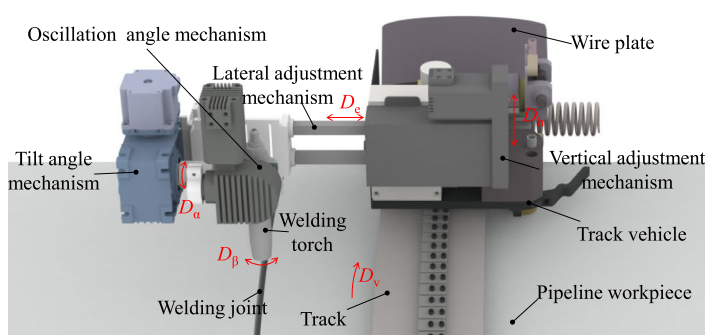
In response to the demand for WTPP control in all-position welding of spatial pipeline butt joint, a 5-DOF pipeline welding robot based on the Pieper criterion is designed firstly, which provides the possibility to achieve the comprehensive control of WTPP during the all-position welding of pipeline at any laying angle. The forward and inverse kinematic models of the designed welding robot are established to provide motion control basis for the adjustment of WTPP. Secondly, a composite sensor is developed, which consists of combined laser structured lights vision sensing and gravity sensing (dual-axis tilt angle sensing). The sensor is integrated with an industrial computer, motion controller, welding power supply, and welding protection gas supplying system to construct a pipeline all-position intelligent welding system. Through a specially designed system control software, real-time automatic control of the WTPP is achieved during the

pipeline all-position welding process, thus achieving high-quality weld formation.

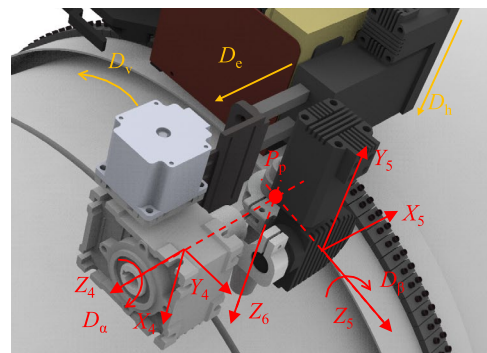
2 Structure and design optimization of 5-DOF pipeline welding robot

To achieve the complete and flexible adjustment requirements of WTPP during the pipeline all-position welding process, a track type 5-DOF pipeline welding robot is designed, as shown in Fig. 2a. It travels on a rigid track through a gear and rack mechanism and has 3 position DOF and 2 posture DOF, including walking DOF D_v , lateral adjustment DOF D_e , vertical adjustment DOF D_h , tilt angle DOF D_α , and oscillation angle DOF D_β . Its main components of each mechanism are listed in Table 1. In addition, Table 2 presents a comparison of the DOF between the designed robot and those described in the previous studies in literatures.

To facilitate the inverse kinematic calculations of the robot, the relationship of the DOF of the welding robot is designed based on the Pieper criterion in kinematics. The axes of the 2 rotational joint axes that control the tilt angle DOF D_α and the oscillation angle DOF D_β are designed as intersection, and the axis of the welding torch also passes through this intersection point, so that the closed form of inverse kinematic solution



(a) Overall schematic diagram of 5-DOF pipeline welding robot



(b) Design of robot end effector DOF based on Pieper criterion

Fig. 2 Mechanism design of welding robot based on Pieper criterion

Table 1 The main composition parameters of each mechanism

Mechanism name (corresponding DOF name)	Model of drive motor	Drive mechanism	Transmission ratio	Joint parameters	Range of motion
Walking mechanism (D_v)	DB40-00330A6-A (servo motor)	Planetary reducer	100:1	θ_1	0 ~ 360°
Vertical adjustment mechanism (D_h)	AM17HDB410-01N (stepping motor)	Ball screw mechanism	50:1	d_2	-40 ~ 40 mm
Lateral adjustment mechanism (D_e)	AM17HDB410-01N (stepping motor)	Ball screw mechanism	50:1	d_3	-40 ~ 40 mm
Tilt angle mechanism (D_α)	AM17HDB410-01N (stepping motor)	Worm gear reducer	50:1	θ_4	-30 ~ 30°
Oscillation angle mechanism (D_β)	AM17HDB410-01N (stepping motor)	Worm gear reducer	50:1	θ_5	-20 ~ 20°

Table 2 Comparison of robotic mechanism degrees of freedom

Mechanism name	Robotic system			
	Our robot	The robot in [8, 12]	The robot in [13]	The robot in [14]
Walking mechanism (D_v)	✓	✓	✓	✓
Vertical adjustment mechanism (D_h)	✓	✓	✓	✓
Lateral adjustment mechanism (D_e)	✓	✓	✓	✓
Tilt angle mechanism (D_α)	✓	×	✓	×
Oscillation angle mechanism (D_β)	✓	×	×	✓

can be obtained. The specific structure design is shown in Fig. 2b. For clear illustration of the 2 rotational axes at the end intersection, the coordinate systems {4} and {5} are translated along the dotted line, and the coordinate origins of {4} and {5} and P_p are actually coincident.

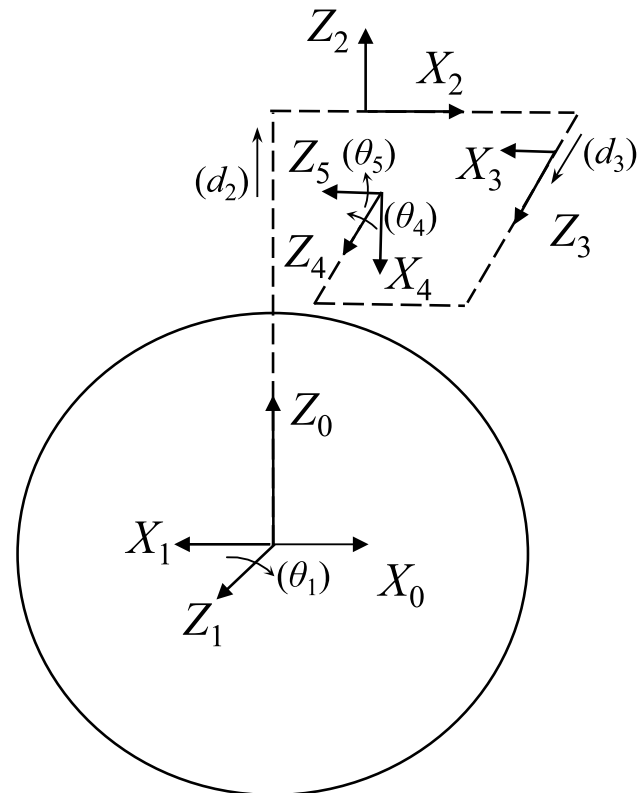
By using online sensing data, the system can perform real-time inverse kinematic calculation and servo control via an industrial computer. This enables adaptive adjustment of WTPP by coordinating the 5 DOF mechanisms, ensuring the welding stability and obtain good weld formation during the pipeline all-position welding process. That is, the intelligent welding of pipeline butt joint can be realized.

3 Kinematic modeling for the 5-DOF welding robot

3.1 Construction of forward kinematic model for the welding robot

Kinematic model and its solution is a prerequisite for trajectory planning and motion control of WTPP. The established D-H model of the designed 5-DOF welding robot using modified Craig D-H method is as shown in Fig. 3. The specific correspondence between the joint DOF and the mechanism is shown in Table 1. In Fig. 3, the origin of coordinate system {4} and {5} are coincided. By establishing the origin coordinates of the end coordinate system at this coincident point, its position coordinates are only affected by the 3 position DOF (D_v , D_h , and D_e), thereby decoupling the calculation of the position parameters and posture parameters of the end coordinate system of the robot, facilitating the calculation and solution of the motion parameters of the robot. The corresponding D-H parameters of each joint of the robot are shown in Table 3.

In Table 3, θ_1 , d_2 , d_3 , θ_4 , and θ_5 are joint variables. Among them, θ_1 is the angle at which the welding robot (track vehicle) travels around the pipeline on track; $D_2 = D/2 + H_0 + H$; D is the diameter of the pipeline; H_0 is the installation height of the track on the pipeline; and H is the adjustment amount of the vertical adjustment mechanism of the welding robot; d_3 is the adjustment amount of the lateral adjustment

**Fig. 3** Schematic diagram of D-H model of 5-DOF pipeline welding robot

mechanism of the welding robot; θ_4 is the tilt angle of the welding torch along the welding direction; θ_5 is the oscillation angle of the welding torch perpendicular to the welding direction; a_1 , a_2 , a_3 , and L_0 are fixed values representing the

Table 3 5-DOF pipeline welding robot D-H parameter table

Joint (i)	a_{i-1} (mm)	α_{i-1} ($^\circ$)	d_i (mm)	θ_i ($^\circ$)
1	0	-90	0	θ_1
2	a_1	90	d_2	180
3	a_2	90	d_3	180
4	a_3	0	L_0	$90 + \theta_4$
5	0	90	0	θ_5

distance or length of the connecting rod between adjacent mechanisms in the designed robot.

In the D-H model, the coordinate transformation between two adjacent linkage coordinate systems is commonly represented by formula (1). By using formula (2) to transform the 5 coordinate systems according to the parameters in Table 3, the forward kinematic expression of the world coordinate

system {0} relative to the end coordinate system {5} in the welding robot can be obtained.

$${}^{i-1}_iT = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_{i-1} \\ \sin\theta_i \cos\alpha_{i-1} & \cos\theta_i \cos\alpha_{i-1} & -\sin\alpha_{i-1} & -d_i \sin\alpha_{i-1} \\ \sin\theta_i \sin\alpha_{i-1} & \cos\theta_i \sin\alpha_{i-1} & \cos\alpha_{i-1} & d_i \cos\alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$${}^0_5T = \prod_{i=1}^5 {}^{i-1}_iT = \begin{bmatrix} -S(\theta_1 + \theta_4)C\theta_5 & S(\theta_1 + \theta_4)S\theta_5 & C(\theta_1 + \theta_4) & (a_1 - a_2 + a_3)C\theta_1 + d_2S\theta_1 \\ S\theta_5 & C\theta_5 & 0 & L_0 + d_3 \\ -C(\theta_1 - \theta_4)C\theta_5 & C(\theta_1 - \theta_4)S\theta_5 & -S(\theta_1 + \theta_4) & -(a_1 - a_2 + a_3)S\theta_1 + d_2C\theta_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

In formula (2), $S\theta$ represents $\sin \theta$; $C\theta$ represents $\cos \theta$; and other variables have similar meanings.

If the distance from the origin of coordinate system {5} to the end P_n of the welding torch tip is l (the clamping length of the welding torch), then its homogeneous coordinates in coordinate system {5} are:

$$P_n = [-l, 0, 0, 1]^T \quad (3)$$

The homogeneous coordinates of the end P_n of the welding torch tip in the world coordinate system {0} can be further derived from formula (2):

$${}^0P_n = {}^0_5TP_n = \begin{bmatrix} l \sin(\theta_1 + \theta_4) \cos\theta_5 + (a_1 - a_2 + a_3) \cos\theta_1 - d_2 \sin\theta_1 \\ -l \sin\theta_5 + (L_0 + d_3) \\ l \cos(\theta_1 - \theta_4) \cos\theta_5 - (a_1 - a_2 + a_3) \sin\theta_1 + d_2 \sin\theta_1 \\ 1 \end{bmatrix} \quad (4)$$

Thus, a forward kinematic model of the designed 5-DOF welding robot has been successfully established.

3.2 Calculation of analytical inverse solution for the welding robot based on Pieper criterion

The inverse kinematic solution methods of closed form solution (analytical solution) include algebraic methods, geometric methods, and transformation methods [24]. The designed 5-DOF welding robot satisfies the Pieper criterion, and its inverse kinematic solution has an analytical solution. In this model of formula (2), since the end coordinate system is established at the intersection point of two rotational axes, its position coordinates are only affected by the first 3 DOF in position, and decoupling of its position and posture parameters can be achieved during inverse kinematic solution. The specific analytical solution solving process is as follows.

In Fig. 4, taking the robot welding on a pipeline as an example, P is the welding point on the circumferential weld of the pipeline butt joint. The welding torch is clamped along the direction of vector O_5P with a length of l , and the coordinate O_5 represents the position parameter of the welding torch. The vector O_5P also represents the posture parameter of the welding torch (the posture angles of the welding torch are α and β).

When solving the inverse kinematic solution, the position and posture matrix of the end point P_n of the welding torch is known as:

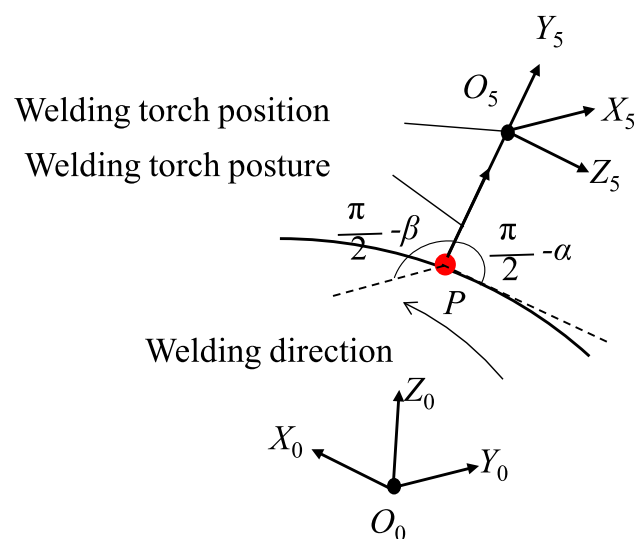


Fig. 4 Schematic diagram of the end coordinate system (welding torch) of the robot

$$T_5 = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

Due to the decoupling of position and posture variables achieved in the structural design, there are only 3 position variables θ_1 , d_2 (H), and d_3 in the last column of the positional coordinates in formula (2). These variables correspond to the known positional coordinates (p_x , p_y , p_z) in formula (5), and the solutions for these 3 position variables can be obtained as follows:

$$\begin{cases} \theta_1 = \arccos\left(-\frac{a_1 - a_2 + d_3}{\sqrt{p_x^2 + p_z^2}}\right) - \arccos\left(\frac{p_x}{\sqrt{p_x^2 + p_z^2}}\right) \\ H = \frac{p_x + (a_1 - a_2 + d_3) \cos(\theta_1)}{\sin(\theta_1)} \\ d_3 = p_y - L_0 \end{cases} \quad (6)$$

For the 2 posture variables θ_4 and θ_5 , they can be calculated using the Euler angle method. The commonly used expression for Euler angle is:

$$Euler(\psi, \theta, \phi) = Rot(z, \psi)Rot(y, \theta)Rot(z, \phi) = \begin{bmatrix} c\psi c\theta c\phi - s\psi s\phi & -c\psi c\theta s\phi - s\psi c\phi & c\psi s\theta & 0 \\ s\psi c\theta c\phi + c\psi s\phi & -s\psi c\theta s\phi + c\psi c\phi & s\psi s\theta & 0 \\ -s\theta c\phi & s\theta s\phi & c\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

If the Euler angle coordinate system $\{E\}$ is established on the lateral displacement mechanism, the homogeneous transformation matrix from coordinate system $\{0\}$ to $\{E\}$ can be expressed as:

$${}^0_E T = {}^0_1 T(\theta_1) {}^1_2 T(H) {}^2_3 T(L) Rot(x, \alpha_3) \quad (8)$$

In formula (8), α_3 is the corresponding connecting rod torsional angle in Table 3.

After calculating ${}^0_E T$, the posture matrix represented by Euler angle can be obtained by combining the posture matrix in ${}^0_E T$ and ${}^0_5 T$.

$$R = {}^0_E R {}^0_5 R \quad (9)$$

By one-to-one correspondence between the variables in formula (9) and the variables in formula (7), the Euler angle variables ψ , θ , and ϕ can be obtained. According to the mechanism design and Euler angle definition, they correspond to the variables θ_4 and θ_5 as follows:

$$\theta_4 = \psi \theta_5 = -\theta \quad (10)$$

By combining formulas (6) to (10), the solutions for the 2 posture variables θ_4 and θ_5 can be obtained as follows:

$$\begin{cases} \theta_4 = -\theta_1 + \arcsin(-a_z) \\ \theta_5 = \arccos(o_y) \end{cases} \quad (11)$$

Thus, the inverse kinematic analytical solution (formulas (6) and (11)) of the 5-DOF robot is obtained. When the welding torch's required positions and postures for the actual welding of the welding point are known, the angles and positions of each joint mechanism of the robot can be obtained based on the above analytical solution, and the rotation angles of each driving motor can be obtained, providing adjustment basis for the real time feedback control of the robot.

4 Construction of pipeline intelligent welding system based on composite sensing

4.1 Composite sensor based on vision sensing

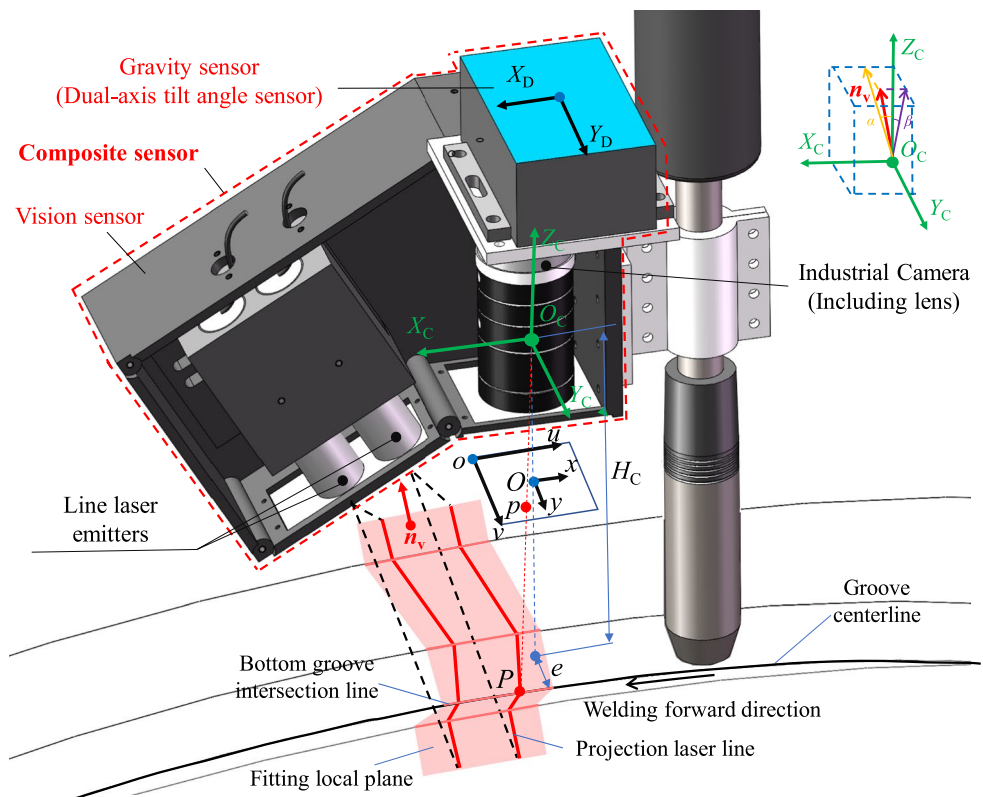
and gravity sensing

The designed composite sensor consists of two parts: a gravity sensor and a vision sensor based on combined line laser structured lights, as shown in Fig. 5 [11].

The vision sensor mainly consists of a monocular industrial camera (including lens and a narrowband filter) and two line laser emitters, and the structure adopts an oblique emitting-direct receiving mode with an angle of 30° , which has a small measurement errors and structural dimensions [25]. During installation, place the composite sensor on the welding forward direction side of the welding torch to detect image and tilt angle data during the welding process. The welding torch axis, camera axis, and two laser emitter axes are coplanar, and the welding torch axis is parallel to the camera axis, so that the welding torch and the sensor have the same posture angle.

During operation, the vision sensor collects two deformed laser lines projected onto the surface of the workpiece to be welded. After capturing the image of the deformed laser lines with the camera, image processing is performed to extract the 2D coordinates of points on the laser lines. Based on the calibrated vision sensor parameters and obtained 2D coordinates of points, using the mathematical detection model based on perspective projection in formula (12), the 3D coordinates of the corresponding points on the deformed laser lines on the

Fig. 5 Schematic diagram of composite sensor composition and detection parameters



surface can be determined in the camera coordinate system [26]. Based on this, welding groove size parameters and relative position and posture parameters of the welding torch can be obtained, providing a basis for adjusting the relative positions of the welding torch. At the same time, by combining the tilt angle collected by the gravity sensor, the absolute postures of the vision sensor can be obtained, and then the absolute postures of the workpiece surface can be obtained, providing a basis for adjusting the welding torch postures that meet the requirement of the welding process parameters.

In Fig. 5, there are 3 rectangular coordinate systems: $O-xy$ is the 2D image coordinate system; $o-uv$ is the 2D pixel coordinate system; and $O_C-X_C Y_C Z_C$ is the 3D camera coordinate system. P is the sampling point on the workpiece surface; p is the corresponding image point in the image plane; and O_C is the optical center of the camera. The detection model formula (12) can achieve the transformation from the 2D coordinates (u, v) of point p to its corresponding 3D coordinates $(X_C, Y_C, \text{ and } Z_C)$ of point P .

$$\begin{cases} Z_C \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X_C \\ Y_C \\ Z_C \\ 1 \end{pmatrix} \\ A_i X_C + B_i Y_C + C_i Z_C + D_i = 0, \quad i = 1, 2. \end{cases} \quad (12)$$

In formula (12), u_0 , v_0 , f_x , and f_y are camera internal parameters; A_i , B_i , C_i , and D_i are the equation parameters of the structured light planes projected by the line laser emitters i ($i = 1, 2$). These two types of parameters are collectively called the vision sensor internal parameters.

4.2 Calculation of welding groove size and WTPP parameters

In the deformed laser line image, the straight-line equation of laser lines can be obtained by straight line fitting. Firstly, the 2D laser line data is segmented according to the intersection points of the obtained laser line image, and then the segmented data is mapped into segmented 3D point cloud according to the detection model formula (12). By plane fitting the corresponding segmented 3D data of the two laser lines, the plane equation of the local groove surface and its end face can be reconstructed, as shown in the fitting local plane in Fig. 5. According to the obtained plane equation of local groove surface and its end face, the groove size parameters such as groove width, depth, groove surface angle, and misalignment value can be calculated [18]. Through calculation, it can be concluded that when applied to pipeline with an outer diameter of 559 mm or more, the error of using the fitting plane instead of local curved surface to solve the sizes of pipeline

groove remains within the 10^{-3} mm range [21], which can be ignored compared with the detection and control accuracy (10^{-1} mm) required for weld formation control in the welding process, so this method can be directly applied without error compensation in this study. Meanwhile, although thermal deformation exists during on-site pipeline external welding, the resulting changes in the groove are generally minor and gradual. The 3D reconstruction process of the local groove surface and its end face is carried out by real-time detection. Therefore, it can effectively overcome the problem of thermal deformation during the welding process, achieve accurate calculation of welding groove sizes, and provide a basis for precise center tracking and control of the welding torch.

The position and posture parameters of the vision sensor (or welding torch) relative to the workpiece include the lateral deviation e (the deviation of the camera optical axis from the groove centerline along the Y_C axis), the relative height H (the distance from the camera focus (welding torch tip) along the optical axis (welding torch axis) to the groove surface), the relative tilt angle α (the angle between the camera optical axis, and the torch axis with respect to the groove end face normal vector $\mathbf{n}_V$ along the welding direction) and the relative oscillation angle β (the angle between the camera optical axis and the torch axis with respect to the groove surface normal vector $\mathbf{n}_V$ perpendicular to the welding direction), as shown in Fig. 5. The calculation formulas for each parameter are as follows:

1. Lateral deviation e

Let l_1 denote the bottom groove intersection line, with a direction vector $\mathbf{n}_1$. The plane passing through l_1 and parallel to the groove end face is denoted as S_p . The coefficients of its four plane equations are A_p , B_p , C_p , and D_p .

Then the deviation e can be expressed as:

$$e = \frac{|\mathbf{MQ} \times \mathbf{n}_1|}{|\mathbf{n}_1|} \quad (13)$$

In formula (13), point M is an arbitrary point on the bottom intersection line l_1 , with coordinates $(X_{CM}, Y_{CM}, \text{ and } Z_{CM})$. Point Q is the intersection of the camera optical axis and the plane S_p , and its coordinates are $(0, 0, -D_p/C_p)$. The vector $\mathbf{MQ}$ represents the directional vector connecting points M and Q .

2. Torch height H

$$H = H_C - H_0 = \left| \frac{D_{pi}}{C_{pi}} \right| - H_0 \quad (14)$$

In formula (14), C_{pi} and D_{pi} are the coefficients of the fitted groove planes S_{pi} intersected by the camera optical

axis. When i takes different values, H_C represents the vertical distance from the vision sensor (camera focal point) to different fitted planes. H_0 is the installation distance between the welding torch tip and the vision sensor (camera focal point). H is the height from the welding torch tip to the groove bottom surface at this detection position.

3. Relative tilt angle α

In the $O_C\text{-}X_C Y_C Z_C$, the normal vector $\mathbf{n}_V$ of the local fitting end face is recorded as $(X_v, Y_v, \text{ and } Z_v)$, then

$$\alpha = \text{sgn}(X_v) \arccos \left| \frac{Z_v}{\sqrt{X_v^2 + Z_v^2}} \right| \quad (15)$$

When α is positive or negative, it indicates the vision sensor (or welding torch) is tilted forward or backward relative to the workpiece surface, respectively.

4. Relative oscillation angle β

Similarly, the angle β can be obtained based on the normal vector $\mathbf{n}_V$, which can be expressed as

$$\beta = \text{sgn}(Y_v) \arccos \left| \frac{Z_v}{\sqrt{Y_v^2 + Z_v^2}} \right| \quad (16)$$

When β is positive or negative, it indicates the vision sensor (or welding torch) oscillates left or right relative to the workpiece surface, respectively.

Therefore, during the pipeline all-position welding process, the position and posture parameters of the vision sensor (or welding torch) relative to the workpiece can be calculated at the current detected position. When the welding torch reaches the detection position, the WTPP can be adjusted according to the planned value, so that the welding torch can achieve accurate welding position and high-quality weld bead formation control.

4.3 Integration of pipeline intelligent welding system

The pipeline intelligent welding system based on composite sensing mainly consists of the 5-DOF pipeline welding robot, the composite sensor, a motion controller (ZMC406 controller), an industrial computer (CPU AMD Ryzen5 5600H, GPU NVIDIA GeForce GTX 1650, RAM 16 GB), and an arc welding system composed of a welding power supply (MEGMEET Artsen Plus 400), a wire feeding subsystem, and a welding

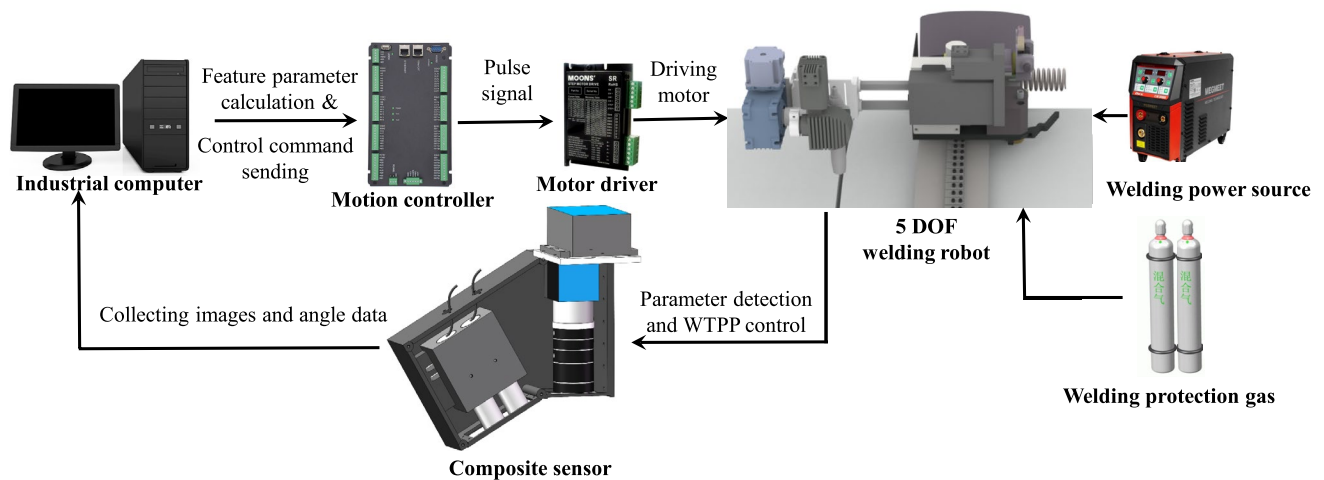


Fig. 6 Composition of pipeline intelligent welding system based on composite sensing and intelligent welding process control

protection gas supply subsystem (a mixture of 20% CO₂ and 80% Ar by volume), as shown in Fig. 6.

During the intelligent welding of pipeline butt joint, the composite sensor collects data (image data and angle data) online. The industrial computer then processes this data using the designed algorithms to solve the welding groove (weld seam) size, posture of the local pipeline to be welded, and the WTPP parameters required. Finally, it sends control instructions to the motion controller. The motor driver controls each joint motor based on the received instructions to achieve real-time control of the welding robot and welding torch. As a result, intelligent WTPP control has been achieved to meet the welding process requirement. A closed-loop real-time control system is constructed and implemented based on online servo feedback on the Visual Studio platform, which is started using a soft trigger method, and the implementation process is shown in Fig. 7.

The constructed pipeline intelligent welding experimental system is shown in Fig. 8. Under the system configuration, the process time of acquisition, transmission, and processing for a single image is about 70 ms, and the acquisition frequency is set to 20 Hz. The real time servo-control of the WTPP can be achieved during the welding process to obtain good weld formation.

5 Simulation analysis of robot workspace and experimental verification

5.1 Analysis of robot motion space based on Monte Carlo method

To verify the rationality of the robot mechanism and the accuracy of the established kinematic model, and judge the

applicability and working ability of the robot, simulation and analysis of the workspace of the designed welding robot end effector structure (welding torch) is conducted based on formula (4) by using Monte Carlo method. The specific implementation process is as follows:

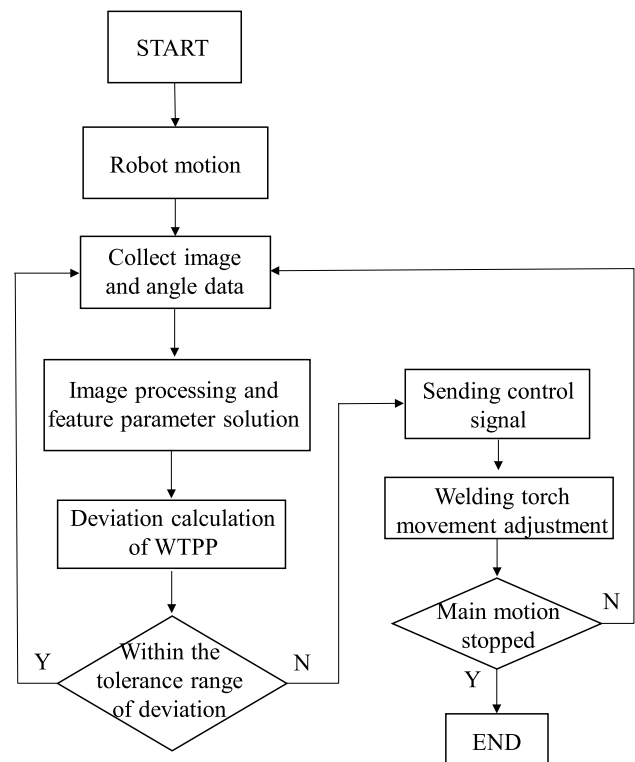


Fig. 7 Real time control flowchart of pipeline intelligent welding based on composite sensing

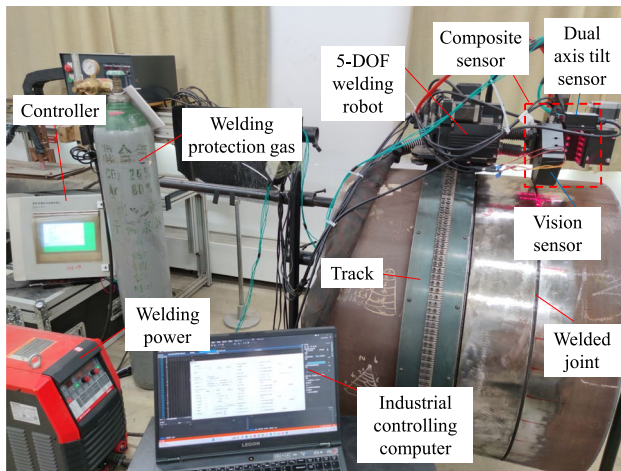


Fig. 8 Pipeline all-position intelligent welding system based on composite sensing

1. The range of values for each variable in Table 3 is $\theta_1 \in [0, 2\pi]$, $d_2 \in [376.5, 446.5]$, $d_3 \in [100, 160]$, $\theta_4 \in [-\pi/6, \pi/6]$, and $\theta_5 \in [-\pi/6, \pi/6]$; the clamping length of the welding torch is 200 mm. When the variables are within this range of values, the mechanical structure operates in accordance with actual needs, and there is no interference among the various structures.
2. In MATLAB, random values of each variable are generated using the *rand()* function, which samples from a

uniform distribution to ensure equal probability across the entire joint range:

$$\theta_{kn} = \theta_i^{\min} + (\theta_i^{\max} - \theta_i^{\min}) \times \text{rand()}. k = 1, 4, 5 \quad (17)$$

$$d_{jn} = d_i^{\min} + (d_i^{\max} - d_i^{\min}) \times \text{rand()}. j = 2, 3 \quad (18)$$

3. Write the values of each variable into the forward kinematic expression (4) in MATLAB to calculate the end effector (welding torch) positions.
4. The simulation was run for 10,000 iterations to generate a dense point cloud representing the 3D workspace of the robot.

Figure 9 is a 3D view of the workspace of the designed 5-DOF pipeline welding robot obtained.

From the simulation results in Fig. 9, it can be seen that when the designed 5-DOF pipeline welding robot moves along the track trajectory, the working space at the end of the welding torch can completely cover the weld seam area to be filled (the position of the pipeline groove), that is, the end of the welding torch can reach any welding point position during the welding process. This indicates that the designed 5-DOF welding robot can meet the welding requirements of pipeline filling and covering and also proves the rationality of the structure design and the accuracy of the established kinematic model.

Fig. 9 3D view of the workspace of the 5-DOF pipeline welding robo

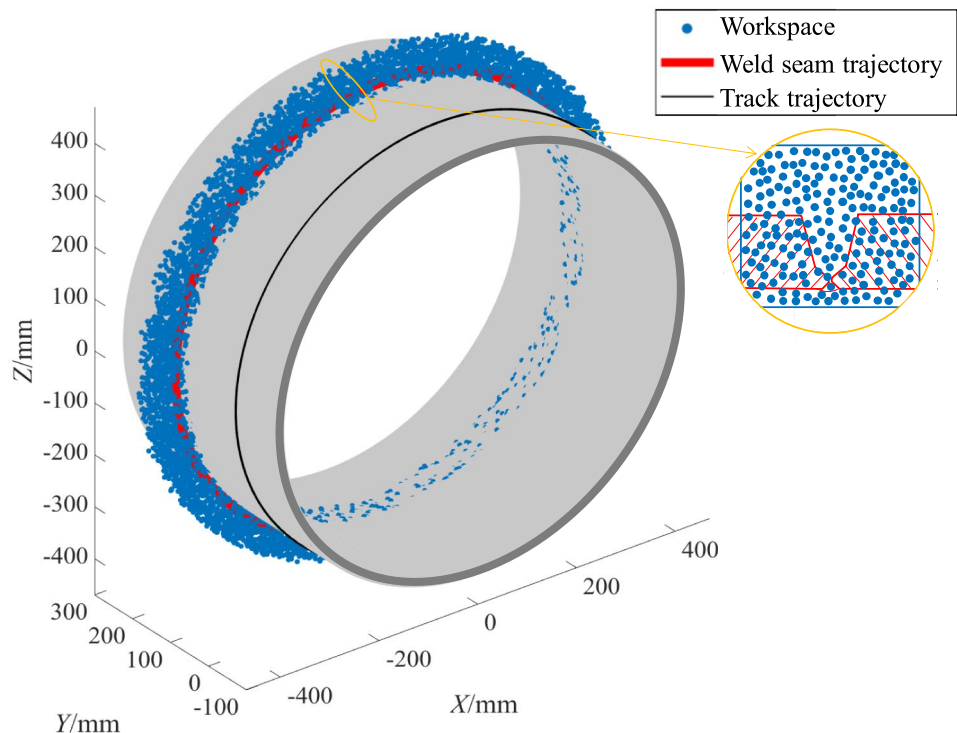


Table 4 Setting of welding experiment process parameters

Parameter	Value
Pipeline diameter	813 mm
Wall thickness	22.73 mm
Groove depth	20.2 mm
Groove width	8.10 mm
Bevel angle	45°/5°
Bottom groove height	2.5 mm
Groove gap	0
Root face	0.8 mm
Steel grade	API 5L X70
Welding wire	ER50-6 (solid wire)
Welding wire diameter	1.0 mm
Welding layer number	5
Welding pass number	6
Welding voltage	25 V
Welding current	190 A
Welding speed	600 mm/min

torch tip end relative to the groove bottom), α (tilt angle of the welding torch relative to the groove bottom), and β (oscillation angle of the welding torch relative to the center of the groove), are combined with the analytical inverse solution to control each joint of the robot. The real-time adjustment curves of the WTPP (e , H , α , and β) are shown in Fig. 11.

In Fig. 11, θ represents the rotation angle of the welding robot around the pipeline; e_v represents the sensing detection value of the deviation between the welding torch and the center of the weld seam; e_c represents the weld seam tracking value achieved by controlling the welding torch based on the sensing detection value e_v ; H_{tv} represents the height of the welding torch tip end relative to the surface of the weld seam; and H_{tc} represents the height tracking value achieved by controlling the welding torch based on the sensing detection value H_{tv} . The average deviation between e_v and e_c was 0.06 mm, and the maximum deviation was 0.25 mm. The average deviation between H_{tv} and H_{tc} was 0.16 mm, and the maximum deviation was 0.63 mm. Meanwhile, during the rotation of the welding robot around the pipeline, the planned values of the welding torch posture angles α and β are α_s and β_s , respectively. From Fig. 11 c and d, it can

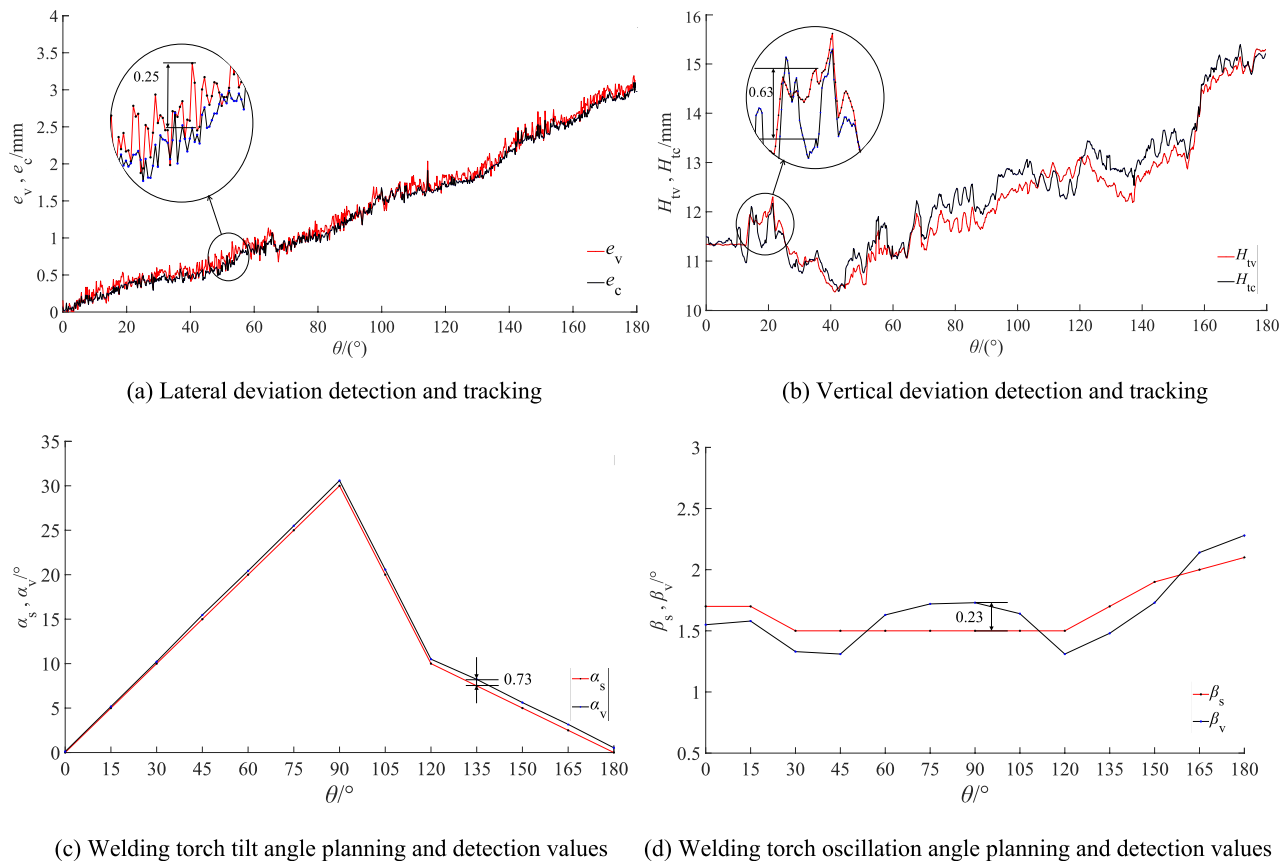
**Fig. 11** Detection and control of WTPP parameters

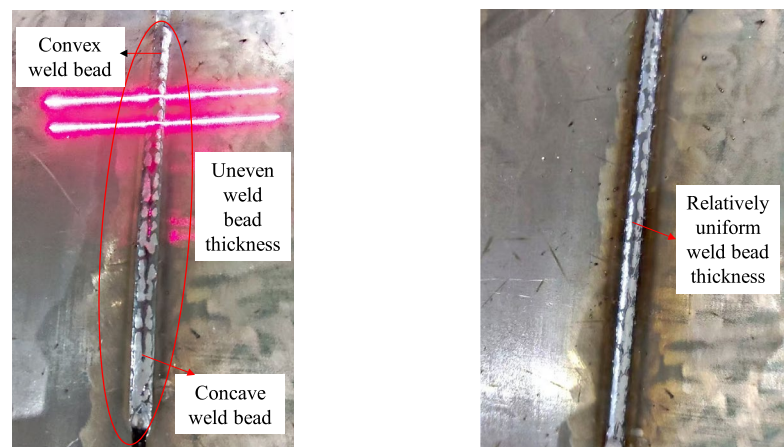
Table 5 Comparison of errors in different weld seam tracking systems

Tracking parameters	Tracking system	Average deviation/mm	Maximum deviation/mm
Lateral deviation	our system	0.06	0.25
	The system in [22]	0.29	0.47
	The system in [23]	0.19	0.53
Vertical deviation	Our system	0.16	0.63
	The system in [28]	0.78	1.75

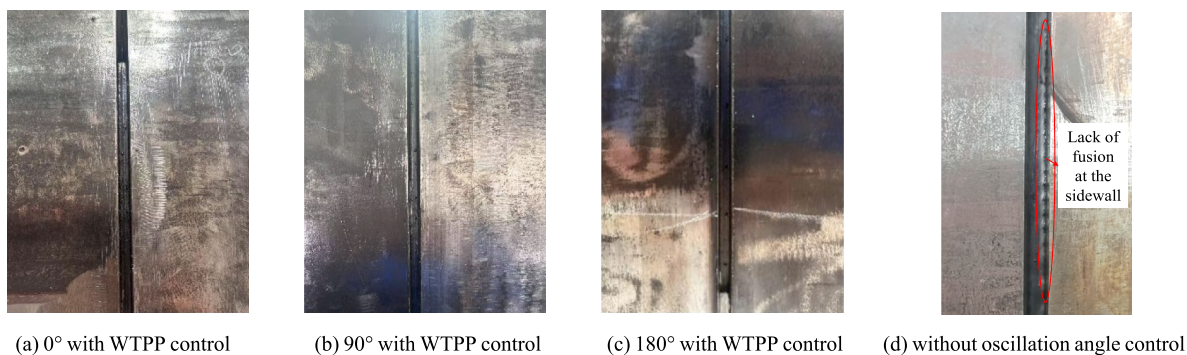
be seen that their maximum deviations from the adjusted detection values α_v and β_v of the welding torch do not exceed 0.73° and 0.23° , respectively. Compared with the required lateral tracking accuracy of 0.4 mm, height tracking accuracy of 1 mm, and posture angle control accuracy of 1° in the usual pipeline welding process, the WTPP control accuracy implemented by the system proves the rationality of the constructed pipeline intelligent welding system and the accuracy of the established kinematic model.

To further demonstrate the advantages of the proposed system, the implemented welding torch lateral and vertical deviation tracking method was compared with seam tracking methods based on deviation calculation from image feature points [22, 23, 28]. As shown in Table 5, the proposed system exhibits better tracking accuracy and stability.

At the same time, during the pipeline all-position welding process, the proposed system can adjust the tilt angle of the welding torch relative to the welding direction, according to the welding process requirements based on the position of the robot at the pipeline, in order to generate a beneficial arc force that partially counteracts the gravity-induced downward flow of the molten pool, thereby improving the consistency of weld bead formation. Compared with the system without the tilt angle adjustment mechanism in references [8, 12, 14], this system achieves a more uniform and consistent weld bead formation by adjusting the tilt angle during pipeline welding. The comparison of weld bead formation near the vertical position is shown in Fig. 12. In addition, the proposed system can dynamically adjust the torch oscillation angle according to the groove size at different

Fig. 12 Weld bead formation effect before and after welding torch tilt angle control near the vertical position

(a) Without welding torch tilt angle control (b) After welding torch tilt angle control

**Fig. 13** Weld bead formation effect before and after comprehensive control of WTPP at pipeline typical positions

welding layers. In this oscillation welding experiment, the torch oscillation speed was set to 1200 mm/min, with a dwell time of 30 ms at both ends, and the torch moved along a zig-zag trajectory. Compared with the system without oscillation angle mechanism in references [8, 12, 13], this system can achieve good fusion effect between the groove sidewalls and the weld bead through the angular oscillation of the welding torch during the welding process, avoiding the occurrence of defects such as lack of fusion. Figure 13 compares the weld formation without oscillation angle adjustment and after WTPP adjustment at typical pipeline positions by the proposed system.

After using the proposed system for intelligent adjustment of the WTPP during the pipeline all-position welding process, the weld bead formation effect of pipeline butt joints in Fig. 13 demonstrates that the weld bead formation at typical positions is accurate, and the weld bead fusion effect is good. This indicates that the designed structure and constructed servo-control system of the pipeline welding robot effectively achieve intelligent adjustment of the WTPP during the pipeline all-position welding process, thereby obtaining high-quality weld formation, laying the foundation for the further development and actual application of intelligent welding of pipeline annular butt joints.

6 Conclusions

1. A 5-DOF welding robot is designed based on the Pieper criterion to meet the demand for all-position welding of pipeline at any laying angle. Its end effector (welding torch) has 3 position control DOF and 2 posture control DOF, which can flexibly achieve WTPP adjustment for any welding position of pipeline annular butt joints to meet the welding process parameter requirement.
2. A forward kinematic model is established for the designed 5-DOF welding robot, and the workspace is simulated and analyzed based on Monte Carlo method to verify the rationality of the robot structure design; The inverse kinematic analytical solution of the robot is obtained for the designed 5-DOF welding robot based on the Pieper criterion, laying the servo-control foundation for flexibly adjusting the WTPP.
3. A pipeline intelligent welding system based on composite sensing is constructed. Based on this, the detection method of welding groove size and WTPP parameters is proposed. Through the designed program control strategy, the intelligent control experiments of the WTPP were carried out during the pipeline all-position welding process. The experiments showed that the tracking deviation of the lateral position between the weld seam center and welding torch was <0.25 mm, the vertical tracking deviation

was <0.63 mm, and the feedback control deviation of the welding torch posture angles were $<0.8^\circ$, which imply that the system can obtain the high-quality weld bead formation and provide support for the improvement of the pipeline intelligent welding level.

Based on the research presented in this study, future work can integrate AI technology to enhance the robustness of image processing, as well as learn from the collected welding torch trajectory data to complete online adaptive trajectory planning for pipelines. This will provide a more intelligent solution for multi-layer multi pass welding of pipelines under various conditions, such as different pipeline laying inclinations and field operation environments.

Author contribution Chuanhui Zhu: investigation, methodology, and writing — original draft. Zihao Wang: methodology, validation, writing — review and editing. Zhiming Zhu: funding acquisition, resources, supervision, writing — review and editing. Jichang Guo: funding acquisition and supervision — review.

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Data availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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